XRST Seminar
A Systematic Literature Review on the Role of Emotions in Virtual Reality

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Abstract. This paper presents a systematic literature review on the role of emotions in virtual reality (VR) experiences, with a specific focus on cooperative emotional games. The review examines the bidirectional relationship between emotional expression and recognition in VR environments and investigates how multimodal physiological data can enhance emotional communication. Two purpose-built VR cooperative games ("Emotion Mirrors" and "Emotion Detective") have been designed to test emotional intelligence skills in virtual environments. These games utilize the Quest Pro's facial expression and eye gaze tracking capabilities alongside heart rate monitoring via P10 (photoplethysmography (PPG) sensor: a non-invasive optical sensor that detects blood volume changes in the microvascular bed of tissue—measures heart rate and HRV by tracking the pulse wave through the skin) to create a comprehensive emotional communication platform. The first game focuses on emotion expression and recognition, including the ability to discern genuine from fake emotions, while the second game explores question-based emotional investigation of visual stimuli. Unlike previous studies that primarily employed single modality measurements or focused on mechanical cooperation, this research creates gameplay mechanics where emotional expression, detection, and authenticity are the central challenges. Previous research by Kim and André (2008) [1] has demonstrated that multimodal physiological measurements can achieve up to 95% accuracy in emotion recognition tasks, significantly outperforming single-channel approaches. Similarly, studies on perspective-taking by Galinsky and Moskowitz (2000) [2] suggest that role reversal activities significantly enhance emotional intelligence development. This research builds on these findings to investigate how VR can be leveraged not only for emotional assessment but also as an active training environment for emotional intelligence skills.

Keywords: Emotional Intelligence · Emotional Authenticity· Cooperative VR Tasks · Heart Rate Monitoring · Facial Expression · Eye Gaze Tracking

1 Introduction

Virtual reality (VR) technology has progressed significantly, creating increasingly immersive environments that can evoke authentic emotional responses [3]. While substantial research has explored emotional reactions in single-user VR settings [4], there remains a critical gap in understanding the nuanced dynamics of emotional communication in cooperative VR environments [5]. This study examines how technological affordances, particularly facial expression tracking and physiological monitoring, can enhance emotional intelligence skills through dedicated cooperative VR experiences [3]. The emotional landscape of human interaction comprises both authentic and deceptive expressions, requiring sophisticated recognition abilities to navigate social situations effectively [6]. Traditional emotional intelligence training faces limitations in providing realistic practice scenarios with immediate feedback [7]. VR offers a unique solution by creating controlled yet immersive environments where emotional expression and recognition can be practiced, measured, and refined [8]. This research introduces two purpose-built cooperative VR games specifically designed to address emotional communication challenges [3]. "Emotion Mirrors" focuses on the expression and recognition of emotions, including the crucial ability to distinguish genuine from fake emotional displays. "Emotion Detective" builds on these skills by requiring players to investigate emotional content through strategic questioning and multimodal feedback interpretation. Both games leverage the Quest Pro's advanced facial expression and eye gaze tracking capabilities alongside heart rate monitoring via P10 sensors to create a comprehensive emotional feedback system. Unlike previous studies that primarily employed passive observation or mechanical cooperation, our approach places emotional expression, detection, and authenticity as the central gameplay mechanics [8]. By analyzing the relationship between physiological signals (heart rate, facial expressions, eye gaze) and emotion recognition accuracy, we aim to identify the most effective feedback mechanisms for enhancing emotional intelligence. Additionally, by implementing role reversal between emotional expresser and detector, we explore how experiencing both sides of emotional communication affects skill development. The findings from this study contribute to both theoretical understanding of emotional dynamics in VR and practical applications for emotional intelligence training. By identifying the relationship between multimodal physiological feedback and emotion recognition accuracy, we establish design principles for more effective emotional training systems. The results have implications for fields including education, therapy, corporate training, and entertainment, where effective emotional communication is important. Prior research provides strong evidence for the effectiveness of multimodal approaches in emotion recognition. Kim and André (2008) achieved 95% recognition accuracy using a combination of four physiological channels (ECG (Electrocardiogram: measures electrical activity of the heart), EMG (electromyography: measures electrical activity in skeletal muscles), skin conductivity (measures changes in the electrical properties of the skin), and respiration (measures breathing patterns: rate, depth, and regularity)), demonstrating that different physiological signals excel at different aspects of emotion

detection—with skin conductivity and EMG correlating strongly with arousal states while ECG and respiration features proving dominant for valence differentiation. This complementary nature of different physiological signals creates a more comprehensive emotional assessment than any single-channel approach [1]. Additionally, the psychological literature on perspective-taking by Galinsky and Moskowitz (2000) [2] and role-based training by Goldstein and Winner (2012) [9]demonstrates that experiencing both sides of emotional communication enhances empathy and emotional intelligence development. Their studies show that individuals who alternate between expressing and detecting emotions develop more nuanced recognition capabilities and improved emotional self-regulation. Our research leverages these established findings to create a purpose-built VR environment specifically designed to amplify these effects through immersive, interactive experiences.

2 Background

2.1 Presence in Virtual Reality

Presence in virtual reality (VR) is a very important concept in VR research; it refers to the feeling of "being there" in a virtual environment, where users feel immersed and engaged as if they were physically present in that space [10]. Several researchers have approached presence from different angles. Some view it as a sense of non-mediation, where we forget about the technology creating our experience. Others see it as a matter of attention and involvement—the more focused we are on the virtual world, the more present we feel in it [10]. The ecological view of presence suggests that when we can naturally interact with a virtual environment, and it responds in expected ways, we develop a stronger sense of being there. Our perception becomes coupled with our possible actions in the virtual space, making the technology itself fade into the background of our awareness. Measuring presence remains challenging because it's primarily a subjective experience. Researchers typically use questionnaires to assess it, though some have explored behavioral and physiological measures. While the relationship between presence and task performance isn't always clear, there's growing evidence that higher presence can amplify emotional responses to virtual stimuli, making it a key factor in applications focused on emotional intelligence and communication [10].

2.2 Presence and Emotions in Virtual Reality

The relationship between presence and emotional experience in virtual reality environments represents an important area of investigation in VR research. Presence is commonly conceptualized as a subjective phenomenon resulting from experiences induced by immersive VR technology [11], and is frequently regarded as a necessary mediator that enables real emotions to be activated within virtual environments [12]. Research examining correlations between presence and emotional experience has produced consistent findings, particularly in VR exposure

therapy contexts. Multiple studies report significant correlations between presence and fear in clinical populations [13], though the relationship appears to be dynamic during exposure to phobic stimuli. For emotions other than fear, findings are more varied. Studies have reported correlations between presence and sadness, joy, and relaxation in healthy controls. The level of immersion also plays a significant role, with higher immersion generally increasing presence [14]. Interestingly, the causal relationship may be bidirectional, as studies manipulating emotional context through information alone (without changing visual stimuli) have demonstrated that emotionally relevant background information enhances presence, suggesting emotions can causally influence presence. The relationship between presence and emotion thus appears to depend both on the specific emotion being studied and individual differences among participants, with arousal potentially serving as a critical mediating factor [14].

2.3 Emotional Intelligence

Emotional intelligence is defined as "a type of social intelligence that involves the ability to monitor one's own and others' emotions, to discriminate among them, and to use the information to guide one's thinking and actions" [15]. The scope of emotional intelligence encompasses three key dimensions: the verbal and nonverbal appraisal and expression of emotion, the regulation of emotion in oneself and others, and the utilization of emotional content in problem-solving contexts [16]. While related to social intelligence, emotional intelligence is more specifically focused on the processing and management of emotional information. This distinction is important for virtual reality applications, as VR environments can be designed to target the development of specific emotional intelligence skills rather than broader social capabilities [16]. In VR contexts, emotional intelligence plays a particularly important role in several ways. It affects how users perceive and respond to emotionally evocative stimuli, influences their ability to recognize and interpret the emotional states of virtual agents or other users, and determines how effectively they can regulate their own emotional responses to potentially intense VR experiences. The immersive nature of VR creates unique opportunities for both assessing and developing emotional intelligence skills through controlled yet realistic emotional scenarios that might be difficult to replicate in physical environments. The relationship between emotional intelligence and presence in VR is bidirectional; higher emotional intelligence may enhance the sense of presence through better emotional engagement with virtual content, while stronger presence experiences may provide more opportunities for exercising and developing emotional intelligence capabilities [16].

2.4 Emotional Authenticity

Emotional authenticity represents a complex philosophical concept central to understanding emotional experiences in VR environments. According to Salmela (2005) [17], authenticity is an important ideal of emotional life, yet determining what constitutes an authentic emotion remains challenging. This challenge is

illustrated by de Sousa's antinomy of emotional authenticity: whether a spontaneous, unreflective emotional reaction is more authentic than a deliberate, reflective response that aligns with one's considered values. In VR contexts, this antinomy becomes particularly relevant, as virtual environments can elicit both immediate emotional responses and more reflective emotional adjustments. The paper [17] distinguishes between sincerity (a psychological concept referring to acknowledging one's actual feelings) and authenticity (a normative notion concerning the alignment between emotions and one's internally justified values and beliefs). For VR experiences focused on emotional communication, this theoretical foundation emphasizes that authenticity involves more than spontaneity—it requires alignment with users' reflectively endorsed values within a coherent pattern of emotional responses that can be recognized as truly their own.

2.5 Heart Rate Analysis for Emotional Authenticity Detection

Heart rate variability serves as a critical physiological indicator for distinguishing between authentic and deceptive emotional expressions in our VR environment. The P10 sensor continuously monitors participants' heart rate, providing real-time data that reveals the autonomic nervous system's response to emotional states and cognitive load associated with deception [1]. Authentic emotional expressions typically produce predictable physiological responses, as documented in extensive psychophysiological research [6]:

- **Fear/Anxiety:** 20-40 BPM increase above baseline (from typical resting HR of 60-100 BPM to 80-140 BPM), reflecting sympathetic nervous system activation and preparation for fight-or-flight response [1]
- **Anger:** 15-30 BPM increase with sustained elevation (from resting 60-100 BPM to 75-130 BPM), indicating arousal and preparation for action, as demonstrated in studies by Levenson et al. (1990) [6]
- **Happiness:** Moderate 5-15 BPM increase (from resting 60-100 BPM to 65-115 BPM), reflecting positive arousal without significant stress, consistent with approach-oriented emotional states [1]
- **Surprise:** Sharp, brief spike of 10-25 BPM (from resting 60-100 BPM to 70-125 BPM) that quickly returns to baseline within 30-60 seconds, reflecting the orienting response to unexpected stimuli [1]

These physiological markers serve as reliable indicators for detecting deception in multiplayer VR environments. Research demonstrates that dishonest acts produce distinct cardiovascular signatures, with lying consistently associated with heightened arousal resulting in increased heart rate acceleration, elevated blood pressure, and slower physiological recovery compared to truth-telling [18]. The involuntary nature of these autonomic responses makes them difficult to consciously control, allowing VR systems to distinguish between genuine emotional expressions and deceptive behaviors. This capability enables more authentic social interactions by providing other players with physiological evidence of emotional authenticity, creating a foundation for trust and meaningful interpersonal connections in virtual environments where traditional social cues may be limited or manipulated [18].

3 Research Question

RQ1: How does real-time physiological feedback (heart rate visualization) enhance the detection of the authenticity of emotional expressions (genuine vs. fake)?
RQ2: How does role reversal in emotional communication (being both expresser and detector) influence emotional intelligence development?

4 Research Requirements

These requirements are based on the literature review and the research questions. The requirements are divided into two categories: technical requirements and emotional requirements. The technical requirements focus on the specific technologies and tools needed to implement the cooperative VR games, while the emotional requirements emphasize the desired emotional outcomes and experiences for users during gameplay. The combination of these two sets of requirements will guide the design and development of the cooperative VR games, ensuring that they effectively elicit and enhance emotional responses in players.

4.1 Technical Requirements

1. **Facial Expression Recognition:** Using the Quest Pro VR headset with built-in face tracking to capture and analyze facial expressions in real-time. This technology enables the detection of subtle emotional cues, enhancing the emotional experience in cooperative VR games.
2. **Eye Gaze Tracking:** Using the Quest Pro VR headset with built-in eye-tracking technology to capture where the player is currently looking in a VR session. This helps us to know which objects in the VR environment are interesting to the players.
3. **Heart Rate Measurement:** Using the P10 sensor to measure heart rate and assess emotional responses during gameplay. This physiological data provides insights into players' emotional states and can be correlated with their in-game experiences.
4. **Visual Emotional Feedback:** Provide real-time visual feedback on players' emotional states based on their physiological measurements. This feedback can help players become more aware of their emotions and how they affect their gameplay.

4.2 Emotional Requirements

1. **Emotional communication direction:** Which direction of emotional communication is developed in the study between the players? Is it one-way or two-way? The emotional communication direction can be either one-way (from the first player to the second player) or two-way (one time from the first player to the second player and another time from the second player to the first player).

2. **Emotional focusing degree:** How much focus is given to the emotional experience of the players? If the focus is on the emotional experience of the players or on the gameplay experience.
3. **Change in Emotional Authenticity:** How authentic are the emotional expressions of the players? Are the players faking their emotional expressions, or are they expressing their true emotions? comparison between helpful and unhelpful behaviors in cooperative VR games.
4. **Emotional intelligence development:** How does the cooperative VR experience contribute to the development of emotional intelligence in the players? Are there any improvements in emotional intelligence skills, such as emotion recognition, empathy, and emotional regulation, as a result of the cooperative gameplay experience?

5 Systematic Literature Review Methodology

This section describes the methods carried out for completing the systematic review of the literature on emotion in virtual reality.

5.1 Search Strategy

The literature search was conducted across the following databases:

- ACM Digital Library
- IEEE Xplore
- ScienceDirect
- Google Scholar

The search query used Boolean operators to combine the following key terms:

- Primary terms: ("virtual reality" OR "VR") AND ("emotion" OR "emotional" OR "affect" OR "affective")
- Secondary terms: AND ("recognition" OR "expression" OR "authenticity" OR "intelligence" OR "physiological" OR "biofeedback" OR "heart rate" OR "facial expression" OR "eye tracking")

5.2 Selection Process and Criteria

The selection process followed a three-stage screening protocol:

1. First screening of titles and abstracts to eliminate obviously unrelated papers.
2. Second screening of abstracts against established eligibility criteria.
3. Full-text reading of remaining paper.

5.3 Data Extraction and Synthesis

Data included from each selected paper included:

- Aims and research questions
- Theoretical framework
- Methodology (participants, apparatus, procedure)
- Consideration of emotional components (recognition, expression, regulation)
- Type of physiological measurements used

A standardized data extraction form (Appendix X) was used to promote consistency. The synthesis stage used thematic analysis to identify patterns that developed across the studies; of particular relevance were studies that considered methods for eliciting affective emotional responses, recognition technologies, and physiological approaches to measurement of emotions.

5.4 Quality Assessment

Each study included was evaluated using a quality assessment framework. Important quality measures included

- Clear research questions/objectives
- Appropriately selected methodology
- Provided relevant technical details of VR implementation
- Provided valid measures for emotional assessments
- Appropriate sample size and participant selection
- Rigorous data analysis

Studies were rated as high, medium, or low quality, with just medium- or high-quality studies remaining in the final review.

6 Related Work

6.1 Effects of Sharing Physiological States of Players in Collaborative Virtual Reality Gameplay

This paper explores how sharing physiological data, especially heart rate, can improve the emotional experience between participants in a cooperative VR game. The researchers implemented for the purpose of this study a novel framework where an active player and an observer player shared the same virtual position but maintained independent viewing orientation. Two distinct VR games were created to elicit different emotional responses: a calm butterfly-catching experience and a tense zombie-shooting scenario. The results showed that the active player's heart rate significantly influenced the observer player's emotional experience, particularly in the tense game [19].

6.2 Embodiment in Virtual Reality Intensifies Emotional Responses to Virtual Stimuli

The project investigates how virtual embodiment influences emotional responses to stimuli in virtual reality environments. The authors hypothesized that experiencing ownership of a virtual body would intensify emotional reactions to virtual stimuli. The results showed that participants who embodied a virtual body reported stronger emotional responses to the stimuli compared to those who did not. The study also found that the intensity of emotional responses was related to the degree of embodiment experienced by participants. This suggests that virtual embodiment can enhance emotional experiences in VR environments, making it a valuable tool for applications such as therapy and training [20].

6.3 Emotional Beasts: Visually Expressing Emotions through Avatars in VR

This work presents "Emotional Beasts," a concept for affective expression in social VR that addresses the lack of emotional expression in current VR avatars. The authors propose a two-fold solution: (1) integrating bio-signal sensors (GSR, PPG) into VR headsets to detect users' emotional states, and (2) generating expressive avatars that visually communicate these emotions. The paper describes a prototype system that combines VR technology, motion capture, and physiological sensing to create emotionally expressive avatars in real-time. Two primary visualization methods are presented: a fur-based system using splines that grow or contract based on emotional states and a particle system that changes color and brightness to reflect emotions. The authors argue this approach moves avatar design away from the uncanny valley toward more expressive representations that can facilitate deeper emotional connections in virtual environments [21].

6.4 Virtual reality facial emotion recognition in social environments: An eye-tracking study

This study investigated a novel VR emotion recognition task for neuropsychiatric assessment and training. Researchers compared emotion recognition across three modalities: VR, photographs, and videos. 100 healthy participants completed all three tasks, with eye-tracking recorded during the VR component. Key findings showed that [22]:

- Recognition accuracy in VR (75% overall) was comparable to photo and video tasks.
- Positive emotions (surprise 93.7%, happiness 86.7%) were recognized most accurately.
- Negative emotions (sadness 69.7%, fear 69.6%, disgust 53.8%) were more difficult to identify.
- Eye-tracking revealed participants focused longer on eyes and nose than mouth areas.

6.5 Emotion Sharing and Augmentation in Cooperative Virtual Reality Games

The work discusses the role of emotions in virtual reality (VR) through facial expression sharing and augmentation in cooperative social VR games. The research presents a prototype system that captures facial expressions from users and shares them between players via avatars, enhancing social presence and communication in virtual environments. The system uses facial motion capture software (F-Clone) and emotion detection (Affective SDK) to map real facial expressions to avatar blend shapes. Two cooperative VR games were implemented to evaluate the system: Bomb Refusal (with asymmetric roles) and Island Survivor (with equal roles requiring negotiation). A pilot study with four participants assessed system usability, avatar positioning preferences, and the overall effectiveness of emotion sharing. Results showed positive usability scores (73.75/100) and revealed that participants preferred in-place avatars over on-screen or adaptive positioning. Interestingly, while avatar expressions were more noticeable in the negotiation-based Island Survivor game, they were considered more useful in the task-oriented Bomb Defusal game, where they served as action warnings [23].

7 Results Review and Evaluation Matrix

Requirements	6.1	6.2	6.3	6.4	6.5
Technical Requirements					
Facial Expression Recognition	×	×	×	✓	✓
Eye Gaze Tracking	×	×	×	✓	✓
Heart Rate Measurement	✓	×	✓	×	✓
Visual Emotional Feedback	✓	✓	✓	✓	✓
Emotional Requirements					
Emotional Communication direction	uni-directional	uni-directional	bi-directional	uni-directional	bi-directional
Emotional Focusing Degree	Medium	high	high	high	Medium
Change in Emotional Authenticity	×	×	×	×	×
Emotional intelligence	×	×	×	✓	✓

Table 1. Evaluation Matrix of Related Works Against Requirements

8 Conception

8.1 General Mini-Games Design

1. **Game 1: Emotion Mirrors (Emotion Recognition)**
   - **Player 1:** The player receives a random emotion in the form of an image and text describing the emotion and has to express it using facial expressions and body language without making any sounds. The heart rate of player 1 is measured using the P10 sensor for three key purposes: (1) to provide physiological validation of emotional authenticity that can't be consciously controlled, allowing player 2 to distinguish between genuine and fake emotional expressions; (2) to create a quantifiable metric for emotional intensity that complements visual cues; and (3) to investigate how physiological feedback impacts the accuracy of emotion recognition

as outlined in RQ1. The facial expression is tracked using the Quest Pro VR headset and visualized on a virtual face. Player 1 can fake the emotion or express it genuinely. Player 1 can also use the eye gaze to look at Player 2 and see if he is looking at him or not.
- **Player 2:** Player 2 has to guess the emotion being expressed by Player 1 based on the facial expressions and body language. The heart rate of player 1 will guide player 2 to guess the emotion. Player 2 can also use the eye gaze to look at Player 1 and see if he is looking at him or not.

2. **Game 2: Emotion Detective**
   - **Player 1:** see a random picture of a person, animal, or object and has to answer player 2's questions about the emotion being expressed in the picture. The heart rate of player 1 is measured using the P10 sensor, and the facial expression is tracked using the Quest Pro VR headset and visualized on a virtual face. Player 1 in this game is the player 2 in the first game.
   - **Player 2:** Player 2 has to ask Player 1 questions about the emotion being expressed in the picture. The heart rate of player 1 will guide player 2 to guess if the emotion is positive or negative.

8.2 Implementation

The implementation of the Emotion Mirrors and Emotion Detective games utilizes Unity 6 as the development platform with the Netcode for GameObjects framework to enable multiplayer and cooperative functionality. Both games follow a client-server architecture where one player (host) serves as the emotion expresser while the other player (client) acts as the emotion detector. The Quest Pro VR headset is used for both players, leveraging its built-in facial expression and eye gaze tracking capabilities. The P10 sensor is integrated to measure heart rate, providing real-time physiological feedback that enhances emotional communication. As shown in figure 1, the Emotion Mirrors game focuses on emotion expression and recognition. Here, player 1 can see the happy emotion photo with a timer and has to express it using facial expressions and body language without making any sounds. Player 2 can see player 1's facial expression and has to guess the emotion being expressed by player 1 based on the facial expressions and heart rate and has to investigate if the emotion is genuine or fake. Player 2 can also see where Player 1 is looking using the eye gaze tracking.

The Emotion Detective game, as shown in figure 2, focuses on emotional investigation through questioning. Player 1 sees a random picture of a person, animal, or object and has to answer player 2's questions about the emotion being expressed in the picture. The heart rate of player 1 is measured using the P10 sensor, and the facial expression is tracked using the Quest Pro VR headset and visualized on a virtual face. Player 1 in this game is the player 2 in the first game.

Also, for each game, the games have a virtual character that copies the player's facial expressions and body language in real-time, providing visual feedback on the player's emotional state. These facial expressions are based on Meta



Fig. 1. Emotion Mirrors Mini Game. Original concept visualization created by the author using Unity.

Face Expressions, which can express all emotions such as happiness, sadness, anger, fear, disgust, surprise, etc. The virtual character's expressions are synchronized with the player's facial movements, enhancing the immersive experience and allowing for more natural emotional communication, as shown in figure 3.

Measuring Emotional Authenticity The measurement of emotional authenticity utilizes a multi-modal approach combining three data streams: heart rate variability (HRV), facial expression congruence, and eye movement patterns. For quantifying emotional authenticity, we analyze the correlation between facial expressions and physiological responses using a custom algorithm that compares (1) the concordance between displayed facial emotion and heart rate patterns typically associated with that emotion based on Levenson's emotion-specific autonomic nervous system activity research [24]; (2) micro-expression leakage indicators detected through the Quest Pro's facial tracking; and (3) gaze behavior patterns.

9 Conclusion

This paper presents a systematic literature review on the role of emotions in virtual reality (VR) experiences, with a specific focus on cooperative emotional games. The review examines the bidirectional relationship between emotional expression and recognition in VR environments and investigates how multimodal physiological data can enhance emotional communication. Two purpose-built VR cooperative games ("Emotion Mirrors" and "Emotion Detective") have been designed to test emotional intelligence skills in virtual environments. These

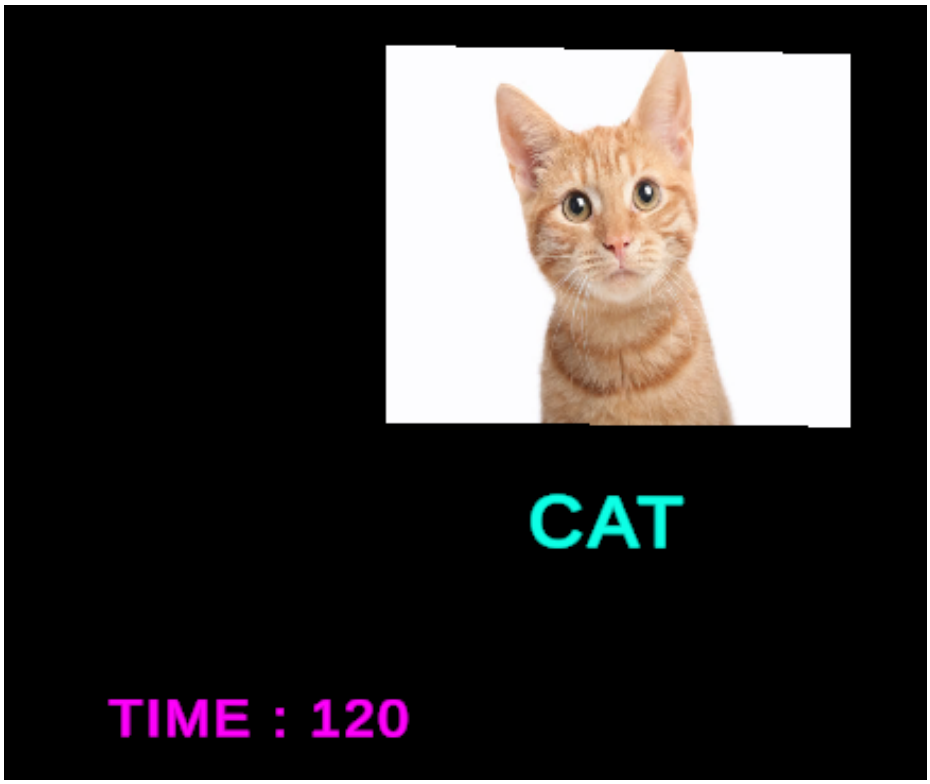


Fig. 2. Emotion Detective Mini Game. Original concept visualization created by the author using Unity.



Fig. 3. Virtual Character Happy and Angry Emotions. Rendered using Meta Face Expressions API in Unity. Character model adapted from Meta's sample assets.

games utilize the Quest Pro's facial expression and eye gaze tracking capabilities alongside heart rate monitoring via P10 sensors to create a comprehensive emotional communication platform. The first game focuses on emotion expression and recognition, including the ability to discern genuine from fake emotions, while the second game explores question-based emotional investigation of visual stimuli. Unlike previous studies that primarily employed single modality measurements or focused on mechanical cooperation, this research creates gameplay mechanics where emotional expression, detection, and authenticity are the central challenges.

9.1 Findings

9.2 Comparative Analysis

10 User Study Questionnaire

10.1 Demographic Questions

1. **Name:** _____
2. **Age:** _____
3. **Gender:** _____

4. **Experience with Emotional Intelligence Training:**
   - ☐ None
   - ☐ Some (workshops, books, courses)
   - ☐ Extensive (professional training, psychology background)

5. **Experience with Social/Multiplayer Games:**
   - ☐ None
   - ☐ Some
   - ☐ Extensive

10.2 Main Questionnaires

Pre-Questionnaire Emotional Intelligence Self-Assessment *Please rate your abilities on a scale of 1-5 (1 = Very Poor, 5 = Excellent):*

1. I can accurately identify my own emotions

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

2. I can recognize emotions in others through facial expressions

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

3. I can detect when someone is expressing fake or inauthentic emotions

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

4. I can effectively express my emotions to others

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

5. I can understand others' emotions through their voice and tone

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

Post-Games Questionnaire Game Experience Evaluation *Please rate your experience on a scale of 1-5:*

1. **Emotional Expression:** How well did you feel you expressed your emotions during the games?

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

2. **Emotional Recognition:** How accurately did you feel you recognized the emotions of your partner?

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

3. **Emotional Authenticity:** How authentic did you feel your partner's emotional expressions were?

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

4. **Physiological Feedback:** How helpful was the heart rate feedback in understanding your partner's emotional state?

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

5. **Face Expression and Eye Gaze Tracking:** How helpful was the face expression and eye gaze tracking in understanding your partner's emotional state?

Very Poorly	Poorly	Average	good	Excellent
1	2	3	4	5

Open-Ended Questions

1. Which emotions were easiest to express/detect? Why?

2. Which emotions were most difficult to express/detect? Why?

3. What strategies did you use to detect fake emotions?

4. What types of questions were most effective in uncovering emotional information?

5. What was the most valuable insight you gained from this experience?

6. How might you apply what you learned about emotional expression and detection in real-world situations?

11 NASA Task Load Index (TLX)

Please rate the following aspects of your experience on a scale from Very Low to Very High:

1. **Mental Demand:** How mentally demanding was the task?

Very Low						Very High
1	2	3	4	5	6	7

2. **Physical Demand:** How physically demanding was the task?

Very Low						Very High
1	2	3	4	5	6	7

3. **Temporal Demand:** How hurried or rushed was the pace of the task?

Very Low						Very High
1	2	3	4	5	6	7

4. **Performance:** How successful were you in accomplishing what you were asked to do?

Very Low						Very High
1	2	3	4	5	6	7

5. **Effort:** How hard did you have to work to accomplish your level of performance?

Very Low						Very High
1	2	3	4	5	6	7

6. **Frustration:** How insecure, discouraged, irritated, stressed, and annoyed were you?

Very Low						Very High
1	2	3	4	5	6	7

12 System Usability Scale (SUS)

Please indicate your level of agreement with each statement:

1. I think that I would like to use this VR system frequently

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

2. I found the VR system unnecessarily complex

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

3. I thought the VR system was easy to use

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

4. I think that I would need the support of a technical person to be able to use this VR system

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

5. I found the various functions in this VR system were well integrated

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

6. I thought there was too much inconsistency in this VR system

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

7. I would imagine that most people would learn to use this VR system very quickly

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

8. I found the VR system very cumbersome to use

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

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